1. Consider a random sample of size $n = 20$ from a normal distribution $X_i \sim N(\mu, \sigma^2)$. Suppose data gave $\bar{x} = 11$ and $s^2 = 16$

i) Test the null hypothesis $H_0: \mu \geq 12$ against the alternative hypothesis $H_1: \mu < 12$ for $\alpha = 0.01$, when it is known that $\sigma^2 = 4$.

$$Z_{calc} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{11 - 12}{2/\sqrt{20}} = \frac{-1\sqrt{20}}{2} = -1\sqrt{5}$$

$$= -2.236 \approx \underbrace{0.0129}$$

since this is greater than our critical value we cannot reject the null hypothesis

ii) What is the Type II error β for the test in i) when in fact $\mu = 10.5$?

get critical value for $\bar{X}$

$$\cancel{-2.236}\ -2.324 = \frac{(\bar{x} - 12)}{2/\sqrt{20}} = -4.\ldots = (\bar{x} - 12)\sqrt{20}$$

$$\bar{x} - 12 = -1.04$$

$$\cancel{\boxed{\bar{x} = 10.96}}$$

$$\bar{x} = 10.96$$

$$P(\bar{X} > 10.96 \mid \mu = 10.5) = P\left[Z > \frac{10.96 - 10.5}{2/\sqrt{20}}\right]$$

$$= P(Z > 1.02859) = 0.1539$$

$$\boxed{\beta = 0.154}$$

iii) What sample size is needed, for the test in (i), in order for the power of the test to be 0.90 for the alternative value $\mu = 10.5$?

?

$\cancel{Z_{0.99}}$ power $= 1 - P(\text{type II})$

$$P\left(Z < \frac{11-10.5}{2/\sqrt{20}}\right) = 0.9$$

$$Z < \frac{-0.5}{2/\sqrt{49}}$$

$$Z < -\frac{\sqrt{n}}{2} = \cancel{(Z < +1.18)} = 1.9$$

$\cancel{0.1355 = 0.9}$

$$Z < -\sqrt{n} = 3.8$$

$$\boxed{n = 14.44\ ?}$$

iv) Test the hypothesis of part (i) where now you assume that σ^2 is unknown.

$$t_{calc} = \frac{(\bar{x} - \mu)}{s/\sqrt{n}}$$

$$\frac{11-12}{4/\sqrt{20}} = \frac{-1\sqrt{20}}{4} = \frac{-2\sqrt{5}}{4} = \frac{-\sqrt{5}}{2}$$

$$= -1.118$$

$df = 20 - 1 = 19$

$t_{19, 0.01} = -2.539$ $> 0.01,$

✩ cannot reject the null hypothesis.

v) Set up the test, i.e., obtain the acceptance/rejection region to test the null hypothesis $H_0: \sigma^2 \leq 9$ against the alternative hypothesis $H_1: \sigma^2 > 9$ with the level of significance $\alpha = 0.01$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

$$= \frac{(20-1)16}{9} = 33.78$$

$$\chi^2_{crit} = 36.19$$

df = 19
$\alpha = 0.01$

* upper tail test
 - we reject the null hypothesis, upper tail test.

2. Let $X_1, \ldots X_{10}$ be a random Sample from a Bernoulli distribution with $Pr(X = 0) = p$. Suppose we want to test the null hypothesis $H_0 : p \le 0.5$ against the alternative hypothesis $H_1 : p > 0.5$. Suppose the rejection region is $R = \{ \Sigma_i x_i \ge 6 \}$

a) Find the power function, & sketch it.

$$B(p) = p\left(\sum_{i=1}^{10} x_i \ge 6 \mid H_1\right)$$

$$B(p) = \sum_{x=6}^{10} \binom{10}{x} p^x (1-p)^{10-x} \qquad p > 0.50$$

b) What is the size of the test?

3. $n=16$ $X_i \sim N(\mu, 1)$ $H_0: \mu = 20$

$\alpha = 0.05$

a) Determine the rejection region of the form

$A = \{\bar{x} \mid -\infty < \bar{x} \leq a\}$

$B = \{\bar{x} \mid b \leq \bar{x} < \infty\}$

lower bound

$-1.645 = \dfrac{\bar{X} - 20}{1/\sqrt{16}} = \bar{X} = 19.589$

$1.645 = \bar{X} = 20.411$ ← upper bound

$-\infty < \bar{X} \leq 19.589$

$20.411 \leq \bar{X} < \infty$